Math 39 Curriculum

Precalculus with Data Analysis | Middlesex School Mathematics | Resource: Math 39

COURSE AT A GLANCE

Linear Modeling and Residuals

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can calculate and interpret average rate of change with appropriate units.

I can construct a linear function from two data points, a table, a graph, or a context.

I can fit a linear regression model to data using technology and interpret its parameters.

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

I can construct an exponential function from two data points, a table, or a context.

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

I can determine and interpret doubling time or half-life from an exponential model.

I can solve exponential and logarithmic equations in modeling contexts.

I can fit an exponential regression model using technology and evaluate its fit.

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can identify a power function and interpret the parameters in y = kx^p.

I can construct a power function from two data points or a proportional relationship.

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

I can recognize and model direct, inverse, and other power-variation relationships.

I can solve equations involving integer or rational powers.

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

I can determine where a combination of functions is positive, zero, or undefined.

I can determine polynomial end behavior from degree and leading coefficient.

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

I can construct a polynomial function from specified zeros, multiplicities, and a point.

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

I can build and optimize a polynomial model in a geometric or economic context.

Describing Data

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can distinguish categorical and quantitative data.

I can construct and interpret frequency and relative-frequency tables.

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

I can identify misleading features in a data display.

I can describe a distribution's modality, symmetry, skewness, and outliers.

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

I can construct and interpret a five-number summary and boxplot.

I can use percentiles and z-scores to locate and compare values within distributions.

I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can organize categorical data in a contingency table or Venn diagram.

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

I can identify an experiment, outcomes, events, and a sample space.

I can calculate theoretical probabilities when outcomes are equally likely.

I can use complements to calculate probabilities.

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can calculate the expected value of a discrete probability situation.

I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

I can linearize power or exponential data and translate between the linearized and original models.